

1.1 SMA 2101: Calculus I

Contact hours: 45

Prerequisites: None

Purpose of the course

To equip the learner with knowledge and skills required for understanding of the laws of calculus and the various methods of differentiation.

Expected Learning Outcomes

By the end of the course, the learner will be able to:

- i. Identify trigonometric functions and their features.
- ii. Differentiate given mathematical equations using first principles.
- iii. Distinguish parametric from implicit higher order derivatives.
- iv. Apply differentiation to practical problems.
- v. Analyze exponential, logarithmic, and hyperbolic functions

Course Content

Limits of functions. Continuity, uniform continuity. Differentiability of functions. Properties of derivatives: Sums, products, quotient, chain rule. Algebraic, logarithms, trigonometric, hyperbolic, and exponential of a single variable. Parametric and implicit differentiation. Higher order derivatives. Applications of derivatives: Small changes, rates of changes, maxima, minima. Curve sketching. Equations of tangent and normal. Kinematics.

Instructional Methods

Lectures (blended), group discussions, presentations.

Instructional Materials and/or Equipment

White boards, white board markers and dusters, computers, projectors.

Student Assessment at Course Level

Continuous assessment (CATs 20% and Assignments 10%) - 30%

End of Semester examination - 70%

Course textbooks

1. Geveci, T. (2015). *Introductory calculus i : Functions, limits, and continuity*. 1st edition. Momentum Press. ISBN: 9781606508534.
2. Goldstein, L. J., Schneider, D. I., Lay, D. C., Asmar, N. H. (2018). *Calculus & Its Applications, Global Edition*. Pearson Education. ISBN: 9781292229102, 1292229101.
3. Stewart, J., Clegg, D. K., Watson, S. (2020). *Multivariable Calculus*. Cengage Learning. ISBN: 9780357042922, 0357042921.
4. Stewart, J. (2012). *Calculus*. Brooks/Cole. ISBN: 9780538498845, 0538498846.

Reference textbooks

1. Geveci, T. (2015). *Intermediate Calculus: Methods of integration*. 1st edition. Momentum Press. ISBN: 9781606508657.
2. Abbott, P. Neill, H. (2010). *Understand Calculus: A Teach Yourself Guide*. McGraw-Hill Education. ISBN: 9780071747608, 0071747605.
3. Hass, J. R., Weir, M. D., Thomas, G. B. (2016). *University Calculus: Early Transcendentals, Global Edition*. Pearson Education. ISBN: 9781292104041, 129210404X.

Course journals

1. Canadian Journal of Mathematics. ISSN: 0008-413X
2. International Journal of Mathematical and Physical Sciences. ISSN: 1307-7465.
3. Journal of Mathematics Research. E-ISSN: 1916-9809.

Reference journals

1. Communications in Mathematical Physics. ISSN: 0010-3616
2. Communications on Pure and Applied Mathematics: ISSN: 1097-0312.
3. Journal of Mathematics. eISSN: 2314-478.